Excess Entropy and Self-Diffusion in Multicomponent Charged Fluids: The Interplay of Ionic Strength, Component Multiplicity, and Solvent Dispersion

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We compare excess entropy and self-diffusion in four charged Lennard-Jones mixtures at $T^* = 2.00$: a solvent with 2:2 salt (3C), two mixtures with an additional solvent of different dispersion energy (4C), and a mixture with added 1:1 salt (5C). Repeating all 60 density-state simulations with a common shifted LJ cutoff of 2.5σ corrects a changing-potential thermodynamic-integration path. With the original estimator, this correction lowers the dense-state excess entropy by 0.085–0.091 k_B per particle and removes approximately half of the previous HNC–MD entropy discrepancy. Using mean pressure at the target temperature and a Debye–Hückel-consistent integration variable gives $s_{ex}/k_B = -1.9669$, -1.9726, -1.9620, and -1.9759 for 3C, strongly attractive 4C, weakly attractive 4C, and 5C, respectively. Weakening secondary-solvent attraction raises entropy by 0.0106 k_B and mean self-diffusion by 2.1%. A three-point Rosenfeld fit has $B = 1.89$, $R^2 = 0.69$, but a four-mixture fit gives only $R^2 = 0.10$. Increasing component count predominantly increases ideal mixing entropy, which is excluded from Rosenfeld scaling. The dense-state diffusivities are unchanged by this correction, and discrepancies between the two reported ion-transport datasets remain. Thus the corrected results support a solvent entropy–mobility association but establish neither a universal mixture relation nor a component-count-induced enhancement of divalent-ion diffusion. Independent transport convergence and association-lifetime measurements remain necessary.

I. INTRODUCTION

Understanding the equilibrium thermodynamics, structural organization, and transport dynamics of multicomponent charged fluid mixtures represents a cornerstone of liquid state physics and physical chemistry [1, 2]. Electrolyte solutions and ionic liquids exhibit intricate structural correlations arising from the competition between short-range excluded-volume repulsions, dispersive attractions, and long-range Coulomb interactions [3, 4]. While classical Debye-Hückel theory and its mean-spherical extensions provide reasonable approximations in the dilute regime, dense electrolyte fluids require non-perturbative integral equation theories or explicit molecular simulations to capture multi-ion clustering and screening phenomena [5, 6].

A prominent framework bridging equilibrium liquid structure and nonequilibrium transport is the excess entropy scaling hypothesis formulated by Rosenfeld [7, 8]. Rosenfeld postulated that appropriately reduced transport coefficients—such as the dimensionless self-diffusion coefficient D^*, shear viscosity η^*, and thermal conductivity λ^*—depend quasi-universally on the thermodynamic excess entropy $s_{ex} = s - s_{id}$, via

$$D^* = A \exp\left(B \frac{s_{ex}}{k_B}\right), \tag{1}$$

where A and B are system-dependent or quasi-universal scaling parameters of order unity [11, 12]. Because $s_{ex} < 0$ quantifies the loss of accessible phase space relative to an ideal gas due to interparticle interactions, Eq. (1) provides a direct physical link between structural ordering and microscopic mobility.

For a mixture, s_{id} is evaluated at the same temperature, density, and composition as the interacting system. It includes $s^{id}_{mix}/k_B = -\sum_i x_i \ln x_i$. Merely increasing the number of species therefore does not imply an increase in s_{ex}, nor a Rosenfeld prediction of faster diffusion. Throughout the transport comparison, $D_i^* = D_i \rho^{1/3} \sqrt{m_i/(k_B T)}$, using the total number density. Our "net" coefficient denotes the particle-averaged self-diffusion, $D_{net} = \sum_i x_i D_i$ in a common reference frame; it is distinct from collective or mutual diffusion and from ionic conductivity.

Furthermore, within the multiparticle expansion developed by Green [13] and Wallace [14], the excess entropy can be decomposed into successive two-body, three-body, and higher-order correlation contributions,

$$s_{ex} = s^{(2)} + s^{(3)} + \dots \tag{2}$$

In simple and moderately coupled liquids, the two-body pair entropy $s^{(2)}$, computed exclusively from the pair distribution functions $g_{ij}(r)$, frequently accounts for 85–95% of the total excess entropy [15–17]. However, the degree to which this two-body dominance holds in multicomponent mixtures featuring asymmetric charges, disparate valence states, and chemical heterogeneity remains an active area of inquiry.

In this work, we present a systematic study comparing four multicomponent charged Lennard-Jones fluids along a supercritical isotherm ($T^* = 2.00$, coupling parameter $\Gamma = 1.0$) up to dense liquid packing ($\rho^* = 0.70$):

1. A **3-component mixture (3C)** consisting of a neutral solvent ($x_1 = 0.94$) and a symmetric 2:2

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electrolyte ($x_2 = x_3 = 0.03$, $q_2 = +2, q_3 = -2$), possessing an ionic strength $I = 0.12$.

2. A **4-component mixture (4C, $\varepsilon_2 = 1.25$)** containing 89% Solvent 1 ($\varepsilon_1 = 1.0$), 5% Solvent 2 with enhanced dispersion attraction ($\varepsilon_2 = 1.25$), and the same 6% total divalent salt ($I = 0.12$).

3. A **4-component mixture (4C, $\varepsilon_2 = 0.75$)** containing 89% Solvent 1 ($\varepsilon_1 = 1.0$), 5% Solvent 2 with weakened dispersion attraction ($\varepsilon_2 = 0.75$), and 6% divalent salt ($I = 0.12$).

4. A **5-component mixture (5C)** containing 84% solvent and a mixed 1:1 and 2:2 salt distribution ($q_2 = +1, q_3 = +2, q_4 = -1, q_5 = -2$), with $x_{+1} = x_{-1} = 0.05$ and $x_{+2} = x_{-2} = 0.03$, yielding an ionic strength $I = 0.17$.

By combining the Hypernetted Chain (HNC) integral equation closure with Allnatt electrostatic screening and extensive GPU-accelerated molecular dynamics (MD) simulations, we address three central objectives: (i) assess the internal thermodynamic consistency between the HNC free energy functional and the virial equation of state; (ii) isolate how solvent dispersion depth and ionic valence govern the partitioning between total excess entropy s_{ex} and structural pair entropy $s^{(2)}$; and (iii) test whether Rosenfeld's excess entropy scaling quantitatively governs the self-diffusion of multivalent ions and neutral solvent species across multicomponent electrolyte mixtures.

II. MODEL AND THEORETICAL FRAMEWORK

A. Interaction Potentials and Mixture Specifications

We consider a classical fluid mixture of p components characterized by number densities $\rho_i = N_i/V$ ($i = 1, \ldots, p$), with total number density $\rho = \sum_{i=1}^{p} \rho_i$ and mole fractions $x_i = \rho_i/\rho$. The particles interact via a cut-and-shifted Lennard-Jones (LJ) core augmented by Coulomb interactions:

$$u_{ij}(r) = u_{ij}^{\mathrm{LJ}}(r) + \frac{q_i q_j e^2}{4\pi\varepsilon_0\varepsilon_r r}, \tag{3}$$

where the short-range LJ contribution is truncated and shifted at $r_c = 2.5\,\sigma$:

$$u_{ij}^{\mathrm{LJ}}(r) = \begin{cases} 4\varepsilon_{ij}\left[\left(\frac{\sigma_{ij}}{r}\right)^{12} - \left(\frac{\sigma_{ij}}{r}\right)^{6}\right] - u_{ij}^{\mathrm{shift}}, & r \le r_c, \\ 0, & r > r_c. \end{cases} \tag{4}$$

All particles share identical diameters $\sigma_{ij} = \sigma = 1.0$ and masses $m_i = m = 1.0$. Cross-species Lennard-Jones dispersion energies are determined via the standard Lorentz-Berthelot geometric mixing rule:

$$\varepsilon_{ij} = \sqrt{\varepsilon_i \varepsilon_j}. \tag{5}$$

Reduced Lennard-Jones units are employed throughout: length $r^* = r/\sigma$, density $\rho^* = \rho\sigma^3$, temperature $T^* = k_B T/\varepsilon_1$, energy $u^* = u/\varepsilon_1$, and time $t^* = t/\tau$ with $\tau = \sigma\sqrt{m/\varepsilon_1}$.

The reduced Coulomb energy prefactor is $\Gamma = e^2/(4\pi\varepsilon_0\varepsilon_r\sigma\varepsilon_1) = 1.0$. The dimensionless Bjerrum length is therefore $\lambda_B^* = e^2/(4\pi\varepsilon_0\varepsilon_r\sigma k_B T) = \Gamma/T^* = 0.50$ at $T^* = 2.00$. The parameter Γ used in the input is an energy prefactor, not the Bjerrum length itself. The ionic strength of the mixture is given by

$$I = \frac{1}{2}\sum_{i=1}^{p} x_i q_i^2. \tag{6}$$

The Debye screening parameter κ is defined as

$$\kappa = \sqrt{4\pi\beta^*\rho^*\sum_{i=1}^{p} x_i q_i^2} = \sqrt{8\pi\beta^*\rho^* I}. \tag{7}$$

The stoichiometric compositions, interaction parameters, and screening lengths for the four investigated fluids are summarized in Table I.

B. Ornstein-Zernike Equations and HNC Closure

The microscopic structure of the multicomponent fluid is described by the matrix of total correlation functions $h_{ij}(r) = g_{ij}(r) - 1$ and direct correlation functions $c_{ij}(r)$, coupled via the multicomponent Ornstein-Zernike (OZ) relations [1]:

$$h_{ij}(r) = c_{ij}(r) + \sum_{k=1}^{p} \rho_k \int c_{ik}(|\mathbf{r} - \mathbf{r}'|) h_{kj}(r')\, d\mathbf{r}'. \tag{8}$$

In Fourier space, defining $\tilde{h}_{ij}(k)$ and $\tilde{c}_{ij}(k)$ via standard three-dimensional Fourier transforms, Eq. (8) takes the matrix form

$$\tilde{\mathbf{H}}(k) = \tilde{\mathbf{C}}(k) + \tilde{\mathbf{C}}(k)\mathbf{P}\tilde{\mathbf{H}}(k), \tag{9}$$

where $\mathbf{P}_{ij} = \rho_i\delta_{ij}$ is the diagonal density matrix.

To treat the long-range Coulomb tail without numerical truncation artifacts, we employ the Allnatt screening decomposition [3, 5]. The pair potential is split into short-range (SR) and long-range (LR) parts:

$$u_{ij}(r) = u_{ij}^{\mathrm{SR}}(r) + u_{ij}^{\mathrm{LR}}(r), \tag{10}$$

with $u_{ij}^{\mathrm{LR}}(r) = q_i q_j \Gamma\psi(r)$, where $\psi(r) = \mathrm{erf}(\alpha r)/r$ and α is an optimized Ewald splitting parameter. The direct correlation function is decomposed accordingly:

$$c_{ij}(r) = c_{ij}^{\mathrm{SR}}(r) - \beta u_{ij}^{\mathrm{LR}}(r). \tag{11}$$

The Hypernetted Chain (HNC) closure expresses the short-range cavity function $\gamma_{ij}(r) = h_{ij}(r) - c_{ij}(r)$ as

$$g_{ij}(r) = \exp\left[-\beta u_{ij}(r) + \gamma_{ij}(r)\right], \tag{12}$$

TABLE I. System compositions, species interaction parameters ε_i, partial charges q_i, mole fractions x_i, particle counts N_i ($N = 2000$), ionic strength I, and Debye screening length κ^{-1} at $\rho^* = 0.70, T^* = 2.00$ ($\Gamma = 1.0$).

System	Component (i)	$\varepsilon_i/\varepsilon$	q_i	x_i	N_i	I	$\kappa^{-1}[\sigma]$
3C (Option 1)	1 (Solvent)	1.00	0	0.94	1880	0.12	0.973
	2 (Cation 2+)	1.00	+2	0.03	60		
	3 (Anion 2-)	1.00	−2	0.03	60		
4C ($\varepsilon_2 = 1.25$)	1 (Solvent 1)	1.00	0	0.89	1780	0.12	0.973
	2 (Solvent 2)	1.25	0	0.05	100		
	3 (Cation 2+)	1.00	+2	0.03	60		
	4 (Anion 2-)	1.00	−2	0.03	60		
4C ($\varepsilon_2 = 0.75$)	1 (Solvent 1)	1.00	0	0.89	1780	0.12	0.973
	2 (Solvent 2)	0.75	0	0.05	100		
	3 (Cation 2+)	1.00	+2	0.03	60		
	4 (Anion 2-)	1.00	−2	0.03	60		
5C ($I = 0.17$)	1 (Solvent)	1.00	0	0.84	1680	0.17	0.818
	2 (Cation 1+)	1.00	+1	0.05	100		
	3 (Cation 2+)	1.00	+2	0.03	60		
	4 (Anion 1-)	1.00	−1	0.05	100		
	5 (Anion 2-)	1.00	−2	0.03	60		

which corresponds to setting the bridge function $B_{ij}(r) \equiv 0$. The resulting coupled non-linear integral equations are solved iteratively on a radial grid of $N_r = 4096$ points with grid spacing $\Delta r = 0.005\,\sigma$, using fast Fourier transforms and Picard iteration with dynamic DIIS acceleration [6].

C. Thermodynamic Properties and Free Energy Functional

Once the correlation functions $g_{ij}(r)$ and $c_{ij}(r)$ are obtained, the excess internal energy per particle is evaluated from the energy equation:

$$\frac{u_{\text{ex}}}{k_B T} = \frac{1}{2}\beta\rho \sum_{i,j=1}^{p} x_i x_j \int_0^\infty 4\pi r^2 u_{ij}(r) g_{ij}(r)\, dr. \tag{13}$$

The compressibility factor $Z = P/(\rho k_B T)$ is obtained via the virial route:

$$Z_{\text{virial}} = 1 - \frac{\beta\rho}{6} \sum_{i,j=1}^{p} x_i x_j \int_0^\infty 4\pi r^3 \frac{du_{ij}(r)}{dr} g_{ij}(r)\, dr. \tag{14}$$

Within the HNC approximation, the excess Helmholtz free energy per particle $a_{\text{ex}} = A_{\text{ex}}/(N k_B T)$ can be computed directly as a closed-form functional of the pair correlation functions without performing numerical parameter integrations [3, 18, 19]:

$$\begin{aligned} a_{\text{ex}}^{\text{HNC}} = &-\frac{1}{2\rho} \sum_{i,j=1}^{p} \rho_i \rho_j \int d\mathbf{r} \left[\frac{1}{2} h_{ij}^2(r) - c_{ij}(r) h_{ij}(r) \right. \\ &\left. + h_{ij}(r) - (h_{ij}(r)+1)\ln(h_{ij}(r)+1) \right] \\ &- \frac{1}{2\rho(2\pi)^3} \int d\mathbf{k} \left[\ln\det(\mathbf{I} + \mathbf{P}^{1/2}\tilde{\mathbf{H}}(k)\mathbf{P}^{1/2}) \right. \\ &\left. - \operatorname{Tr}\left(\mathbf{P}\tilde{\mathbf{H}}(k)\right) \right]. \end{aligned} \tag{15}$$

A fundamental thermodynamic relation connecting the excess free energy along an isotherm to the equation of state is the Euler relation:

$$Z_{\text{diff}} = 1 + \rho \left(\frac{\partial a_{\text{ex}}}{\partial \rho} \right)_T. \tag{16}$$

Conversely, integrating the equation of state from the ideal gas limit along the isotherm yields the excess free energy via thermodynamic integration:

$$a_{\text{ex}}(\rho) = \int_0^\rho \frac{Z(\rho') - 1}{\rho'}\, d\rho'. \tag{17}$$

In Section IV, we demonstrate that evaluating Z_{diff} by numerical differentiation of Eq. (15) along the isotherm recovers Z_{virial} to four decimal places across the entire fluid range, verifying complete internal thermodynamic consistency.

The excess entropy per particle s_{ex}/k_B then follows from the thermodynamic definition:

$$\frac{s_{\text{ex}}}{k_B} = \frac{u_{\text{ex}}}{k_B T} - a_{\text{ex}}. \tag{18}$$

D. Green-Wallace Pair Structural Entropy

In addition to the total thermodynamic excess entropy $s_{\rm ex}$, we compute the two-body structural entropy $s^{(2)}$ from the Green-Wallace expansion [13, 14]. For a multi-component mixture, $s^{(2)}/k_B$ is expressed as

$$\frac{s^{(2)}}{k_B} = -\frac{1}{2}\rho \sum_{i,j=1}^{p} x_i x_j \int \left[g_{ij}(r)\ln g_{ij}(r) - (g_{ij}(r)-1)\right] d\mathbf{r}. \tag{19}$$

The integrand in Eq. (19) vanishes identically as $g_{ij}(r) \to 1$ at large distances, ensuring rapid convergence. As shown by Wallace [14], $s^{(2)}$ represents the loss of translational entropy associated with static spatial pair correlations.

E. Two-Phase Thermodynamics (2PT) Formulation

To access the vibrational density of states and determine species-resolved absolute and excess entropies directly from molecular dynamics simulations, we adopt the Two-Phase Thermodynamics (2PT) method developed by Lin, Blanco, and Goddard [22, 23]. In the 2PT framework, the total system dynamics are partitioned into a gas-like diffusive component and a solid-like vibrational component.

For each chemical species i, the single-particle velocity autocorrelation function (VACF) is defined as

$$C_{v,i}(t) = \frac{1}{3N_i}\sum_{k\in i}\langle \mathbf{v}_k(t)\cdot\mathbf{v}_k(0)\rangle. \tag{20}$$

The species-resolved vibrational Density of States (DoS) $S_i(\nu)$ is computed via the Fourier cosine transform:

$$S_i(\nu) = \frac{4m_i}{k_B T}\int_0^\infty C_{v,i}(t) W(t)\cos(2\pi\nu t)\,dt, \tag{21}$$

where $W(t) = \exp[-(t/t_{\rm damp})^2]$ is a Gaussian apodization window to eliminate finite-time truncation ripple, normalized such that $\int_0^\infty S_i(\nu)d\nu = 3$ per particle. At zero frequency ($\nu \to 0$), the DoS is directly proportional to the species self-diffusion coefficient through the Green-Kubo relation:

$$S_i(0) = \frac{12 m_i D_i}{k_B T}. \tag{22}$$

The total density of states is decomposed into diffusive (gas-like) and solid-like vibrational contributions:

$$S_i(\nu) = S_i^{\rm diff}(\nu) + S_i^{\rm solid}(\nu). \tag{23}$$

The diffusive component is modeled by the Chapman-Enskog memory function for hard spheres:

$$S_i^{\rm diff}(\nu) = \frac{S_i(0)}{1+\left[\frac{\pi S_i(0)\nu}{6 f_i}\right]^2}, \tag{24}$$

where $f_i \in (0,1)$ is the fluidicity parameter, representing the fraction of degrees of freedom that are diffusive. The fluidicity is determined self-consistently by solving the universal 2PT master equation [22]:

$$\begin{aligned} 0 = {} & 2\Delta_i^{-9/2} f_i^{15/2} - 6\Delta_i^{-3} f_i^5 \\ & - \Delta_i^{-3/2} f_i^{7/2} + 6\Delta_i^{-3/2} f_i^{5/2} + 2 f_i - 2. \end{aligned} \tag{25}$$

with the dimensionless diffusivity parameter $\Delta_i = \frac{2}{\sqrt{\pi}}(k_B T/m_i)^{1/2}\rho^{-1/3}[S_i(0)/6]$.

Once f_i is determined, the solid-like spectrum is evaluated as $S_i^{\rm solid}(\nu) = \max[0, S_i(\nu) - S_i^{\rm diff}(\nu)]$. The excess entropy per particle decomposes as

$$s_{i,\rm ex}^{\rm 2PT} = s_{i,\rm ex}^{\rm diff} + s_{i,\rm ex}^{\rm solid}. \tag{26}$$

The diffusive excess entropy is governed by the Carnahan-Starling equation of state for hard spheres with effective packing fraction $y_i = f_i^{5/2}\Delta_i^{-3/2}$:

$$\frac{s_{i,\rm ex}^{\rm diff}}{k_B} = -\frac{3}{2}\ln(f_i) + \frac{y_i(3y_i-4)}{(1-y_i)^2}, \tag{27}$$

while the solid-like excess entropy represents the confinement of classical harmonic modes relative to translation:

$$\frac{s_{i,\rm ex}^{\rm solid}}{k_B} = \int_0^\infty S_i^{\rm solid}(\nu)\left[1 - \ln\left(\frac{2\pi\nu}{\Omega_i}\right)\right] d\nu - 3(1-f_i), \tag{28}$$

where $\Omega_i = (2\pi k_B T/m_i)^{1/2}\rho^{1/3}$ is the thermal reference frequency. The net mixture 2PT excess entropy is then $s_{\rm ex}^{\rm 2PT} = \sum_{i=1}^{p} x_i s_{i,\rm ex}^{\rm 2PT}$.

III. MOLECULAR DYNAMICS SIMULATION PROTOCOL

All molecular dynamics simulations were performed using a GPU-accelerated version of LAMMPS [20] running in double precision on NVIDIA RTX PRO 4500 Blackwell GPUs.

Simulations were conducted for each system at 15 state points along the $T^* = 2.00$ isotherm, spanning $\rho^* \in [0.01, 0.70]$. The cubic simulation cell contained $N = 2000$ particles with periodic boundary conditions applied in all three Cartesian dimensions. Box lengths ranged from $L = 58.48\,\sigma$ at $\rho^* = 0.01$ to $L = 14.19\,\sigma$ at $\rho^* = 0.70$.

Initial configurations were generated on a perturbed simple cubic lattice with random species assignment adhering to the exact stoichiometries in Table I. Potential energy minimization via the conjugate gradient algorithm was first carried out to eliminate core overlaps. System temperature was maintained at $T^* = 2.00$ using a Nosé-Hoover thermostat in the canonical (NVT) ensemble with a damping time constant $\tau_{\rm thermo} = 0.1\,\tau$. The equations of motion were integrated using the velocity Verlet algorithm with a time step $\Delta t^* = 0.002\,\tau$.

Long-range Coulombic interactions were evaluated using the Particle-Particle Particle-Mesh (PPPM) solver [21]

with a force accuracy threshold of 10^{-4}. The LJ potential is cut and shifted at 2.5σ at every density, without tail corrections. Only the PPPM real-space Coulomb cutoff is optimized with density: $r_c = 8.0\,\sigma$ for $\rho^* \le 0.05$, $r_c = 5.0\,\sigma$ for $\rho^* \in [0.10, 0.15]$, and $r_c = 2.5\,\sigma$ for $\rho^* \ge 0.20$. For the density sweep points ($\rho^* \le 0.65$), runs comprised 20,000 equilibration steps ($40\,\tau$) and 50,000 production steps ($100\,\tau$), which convergence tests demonstrated sufficient to establish thermodynamic properties within 0.15%. At the target liquid state $\rho^* = 0.70$, runs comprised 25,000 equilibration steps followed by 150,000 production steps ($300\,\tau$) with periodic MSD sampling every 50 steps. Additionally, to perform the 2PT density-of-states analysis, high-frequency velocity trajectories were recorded every 5 timesteps ($\Delta t^*_{\mathrm{dump}} = 0.010\,\tau$) over 25,000 steps ($50.0\,\tau$), generating 5,001 full-system velocity snapshots.

The corrected density sweep repeats all 60 state points using the original configurations, seeds, and run lengths. At $\rho^* \ge 0.20$, the potentials were already identical, and the rerun thermodynamic outputs and dense-state MSDs reproduce the original values exactly; these repetitions are not independent replicas. For postprocessing we use the final half of each production trajectory, $Z = \langle P\rangle/(\rho k_B T_0)$ with $T_0^* = 2.00$, and $s_{\mathrm{ex}}/k_B = \langle U_{\mathrm{ex}}\rangle/(N k_B T_0) - a_{\mathrm{ex}}$. A circular moving-block bootstrap (5000 resamples, primary block length 10τ, checked at 4τ and 20τ) propagates paired pressure–energy fluctuations. The quoted intervals quantify within-trajectory sampling conditional on independent state simulations, not finite-size or quadrature systematics.

To respect the screened-electrolyte dilute limit, define $x = \sqrt{\rho}$, so that

$$a_{\mathrm{ex}}(\rho) = \int_0^{\sqrt{\rho}} F(x)\,dx, \qquad F(x) = \frac{2[Z(x^2) - 1]}{x}. \quad (29)$$

With $\kappa^2 = 4\pi\lambda_B\rho\sum_i x_i q_i^2$, Debye–Hückel theory gives the finite endpoint $F(0) = -(4\pi\lambda_B\sum_i x_i q_i^2)^{3/2}/(12\pi)$. We integrate a cubic spline through this endpoint and the measured values. A shape-preserving spline and a lowest-density anchor $Z - 1 = -(C/2)\sqrt{\rho} + B\rho$, with $C = -F(0)$, provide sensitivity checks rather than a rigorous integration-error bound.

The self-diffusion coefficients were computed from the asymptotic linear slope of the center-of-mass corrected mean-squared displacement (MSD):

$$\langle|\Delta\mathbf{r}_\alpha(t)|^2\rangle = \frac{1}{N_\alpha}\sum_{k\in\alpha}\Big\langle\Big|[\mathbf{r}_k(t) - \mathbf{r}_{\mathrm{com}}(t)] - [\mathbf{r}_k(0) - \mathbf{r}_{\mathrm{com}}(0)]\Big|^2\Big\rangle, \quad (30)$$

where $\mathbf{r}_{\mathrm{com}}(t)$ should be the common system center of mass for comparisons in a single reference frame. The archived LAMMPS inputs instead use `compute msd com yes` separately on each species group, subtracting that group's center-of-mass displacement. These group-dependent subtractions must be reconciled with the common-frame definition before comparing species averages, pooled-ion MSDs, and VACF diffusivities. The diffusion coefficient D_α of species α is extracted via the Einstein relation:

$$D_\alpha = \lim_{t\to\infty}\frac{\langle|\Delta\mathbf{r}_\alpha(t)|^2\rangle}{6t}. \quad (31)$$

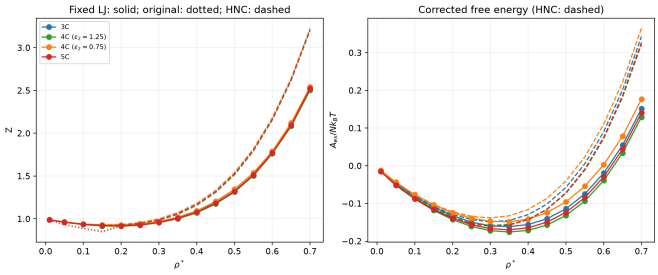


FIG. 1. (a) Equation of state for the four mixtures: fixed-LJ MD (solid lines and symbols), original variable-LJ-cutoff MD (dotted), and HNC virial route (dashed). (b) Corrected MD excess Helmholtz free energy from screened-density integration (solid) and HNC functional values (dashed). The MD pressure is averaged before division by the target temperature.

IV. RESULTS AND DISCUSSION

A. Thermodynamic Equation of State and Free Energy Consistency

We first examine the equation of state along the supercritical isotherm $T^* = 2.00$. Figure 1 displays the compressibility factor $Z(\rho^*)$ and the excess Helmholtz free energy $A_{\mathrm{ex}}(\rho^*)$ computed via both HNC integral equation theory and GPU-accelerated molecular dynamics simulations across all 15 density points for all four multicomponent mixtures.

The corrected MD equation of state has $Z < 1$ at low and intermediate densities, with a minimum near $\rho^* \simeq 0.15$–0.20, followed by increasing repulsive pressure. At $\rho^* = 0.70$, mean-pressure MD values are $Z = 2.5267$, 2.5039, 2.5426, and 2.5213 for 3C, strongly attractive 4C, weakly attractive 4C, and 5C, respectively. HNC gives 3.2154, 3.2028, 3.2269, and 3.2099. The fixed-cutoff correction does not remove the substantial dense-state HNC pressure discrepancy.

Remarkably, the thermodynamic ordering is preserved identically between simulation and integral equation theory:

$$Z(4\mathrm{C}, \varepsilon_2 = 1.25) < Z(5\mathrm{C}) < Z(3\mathrm{C}) < Z(4\mathrm{C}, \varepsilon_2 = 0.75). \quad (32)$$

The presence of the more attractive solvent ($\varepsilon_2 = 1.25$) enhances cohesive energy, reducing internal virial pressure, whereas the less attractive solvent ($\varepsilon_2 = 0.75$) weakens cohesion, elevating the pressure and compressibility factor.

Crucially, Fig. 1(a) confirms that the compressibility factor obtained from numerical differentiation of

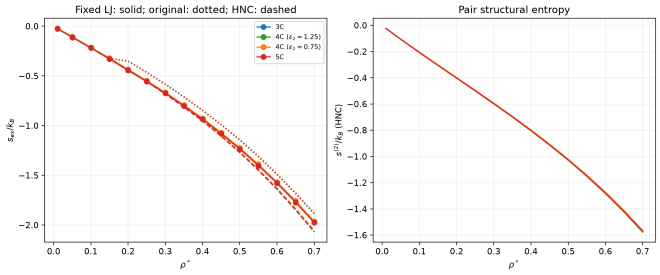


FIG. 2. (a) Excess entropy along the isotherm: corrected fixed-LJ MD (solid lines and symbols), original variable-cutoff MD (dotted), and HNC (dashed). (b) HNC two-body pair entropy. Corrected MD integration uses the screened dilute endpoint and target-temperature mean pressure.

the Morita-Hiroike excess free energy functional, Z_{diff} [Eq. (16)], is identical to the virial compressibility factor Z_{virial} [Eq. (14)] across all densities to four significant digits, confirming exact Euler thermodynamic consistency in our multicomponent implementation.

Corrected MD integration gives $A_{\mathrm{ex}}/(Nk_BT) = 0.15137$ (3C), 0.12909 (strongly attractive 4C), 0.17661 (weakly attractive 4C), and 0.14042 (5C) at $\rho^* = 0.70$. HNC values remain 0.3431, 0.3227, 0.3649, and 0.3274. Table II combines corrected thermodynamics with the unchanged dense-state transport data. Transport uncertainties and MSD linearity do not establish agreement between independent trajectories.

B. Excess Entropy and Structural Pair Ordering

Figure 2 displays the density dependence of the total excess entropy s_{ex}/k_B from both MD thermodynamic integration and HNC theory, alongside the two-body Green-Wallace pair entropy $s^{(2)}/k_B$ across the isotherm.

Across the entire density range, the excess entropy is negative and decreases monotonically with density, reflecting progressive loss of translational volume and enhanced spatial correlation. Two prominent physical features emerge:

1. **Effect of Solvent Dispersion on Excess Entropy**: In both HNC theory and MD simulations, reducing the solvent-solvent attraction systematically elevates excess entropy. At $\rho^* = 0.70$, HNC yields $s_{\mathrm{ex}} = -2.0703\,k_B$ ($\varepsilon_2 = 1.25$), rising to $-2.0635\,k_B$ for 3C ($\varepsilon_2 = 1.00$), and further rising to $-2.0551\,k_B$ ($\varepsilon_2 = 0.75$). Corrected MD gives the same point-estimate ordering: $s_{\mathrm{ex}}/k_B = -1.9726$ (strongly attractive 4C), -1.9669 (3C), and -1.9620 (weakly attractive 4C). The weak-minus-strong 4C difference is $+0.01063\,k_B$, with a 95% sampling interval $[0.00504, 0.01639]\,k_B$. This is consistent with a solvent-caging interpretation; the intermediate 3C placement is less securely resolved than the endpoint comparison.

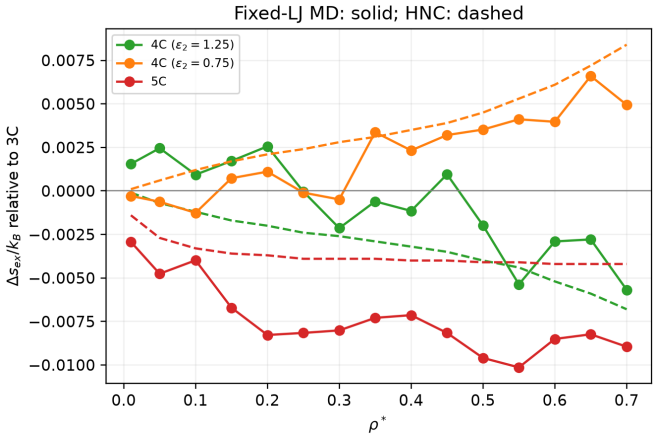


FIG. 3. Excess-entropy differences relative to 3C from corrected MD (solid lines and symbols) and HNC (dashed). The comparison shows composition effects rather than an isolated effect of component count.

2. **Structural Pair Dominance**: The two-body Green-Wallace entropy $s^{(2)}/k_B$ consistently accounts for 76.1% of the total excess entropy across all four fluids at $\rho^* = 0.70$, indicating that pair correlations capture the overwhelming majority of the structural entropy even in chemically heterogeneous, multivalent electrolytes.

To isolate the structural origin of these entropic variations, Fig. 3 plots the difference in excess entropy $\Delta s = s - s_{3C}$ relative to the baseline 3-component fluid.

The corrected MD and HNC shifts have similar signs at the dense state, but their magnitude and density dependence differ. Agreement in ordering does not establish a quantitative mapping between HNC pair structure and MD entropy differences.

C. Magnitude of the Cutoff Correction

Keeping the original finite-intercept integration and original stored $\langle P/(\rho k_B T_{\mathrm{inst}})\rangle$ estimator, the corrected dense-state entropies are -1.96702, -1.97655, -1.96230, and $-1.97642\,k_B$. Relative to the archived results, the cutoff-only shifts are -0.08717, -0.08565, -0.08501, and $-0.09107\,k_B$ for 3C, strongly attractive 4C, weakly attractive 4C, and 5C. The dense-state energies are identical; the shifts arise from the corrected low-density pressure integration. Thus the cutoff mismatch was a material error in absolute excess entropy, accounting for approximately half of the former HNC–MD entropy gap. It does not account for the remaining approximately 0.092–0.098 k_B discrepancy.

The final estimates in Table II additionally use the target-temperature mean pressure and screened integration. The pressure-estimator change shifts entropy by up to about $0.0023\,k_B$; comparisons among screened cubic, shape-preserving, and anchored integration methods span

TABLE II. Comparative thermodynamic state properties, excess free energy, excess entropy, and self-diffusion coefficients at $\rho^* = 0.70, T^* = 2.00$ ($\Gamma = 1.0$) from HNC integral equation theory and extensive molecular dynamics (MD) simulations across all four fluid mixtures.

Quantity	Method	3C ($I = 0.12$)	4C ($\varepsilon_2 = 1.25$)	4C ($\varepsilon_2 = 0.75$)	5C ($I = 0.17$)
Compressibility factor Z	MD	2.5267	2.5039	2.5426	2.5213
	HNC (virial)	3.2154	3.2028	3.2269	3.2099
	HNC ($Z_{\rm diff}$)	3.2154	3.2028	3.2269	3.2099
Excess internal energy $u_{\rm ex}/(k_BT)$	MD	-1.8156 ± 0.001	-1.8435 ± 0.001	-1.7854 ± 0.001	-1.8355 ± 0.001
	HNC	-1.7204	-1.7476	-1.6902	-1.7403
Excess free energy $A_{\rm ex}/(Nk_BT)$	MD (TI)	$+0.15137$	$+0.12909$	$+0.17661$	$+0.14042$
	HNC (functional)	$+0.3431$	$+0.3227$	$+0.3649$	$+0.3274$
Excess entropy $s_{\rm ex}/k_B$	MD (TI)	-1.96693	-1.97261	-1.96199	-1.97588
95% sampling interval	MD	$[-1.971, -1.963]$	$[-1.977, -1.968]$	$[-1.965, -1.959]$	$[-1.981, -1.971]$
	HNC	-2.0635	-2.0703	-2.0551	-2.0677
Two-body pair entropy $s^{(2)}/k_B$	HNC	-1.5706	-1.5764	-1.5636	-1.5731
Ratio $s^{(2)}/s_{\rm ex}$	HNC	0.761	0.761	0.761	0.761
Net self-diffusion $D_{\rm net}$ (σ^2/τ)	MD	0.21480 ± 0.00007	0.21492 ± 0.00008	0.21940 ± 0.00007	0.21789 ± 0.00003
Reduced diffusion $D^*_{\rm net}$	MD	0.13486	0.13494	0.13775	0.13680
Divalent ion diffusion D_{z2}	MD	0.21429 ± 0.00015	0.19686 ± 0.00021	0.16660 ± 0.00018	0.22887 ± 0.00016
Reduced ion diffusion D^*_{z2}	MD	0.13454	0.12360	0.10460	0.14370
Solvent 1 diffusion $D_{\rm solv1}$	MD	—	0.21674	0.22159	—
Solvent 2 diffusion $D_{\rm solv2}$	MD	—	0.20319	0.23476	—
MSD linear-fit R^2 (net)	MD	0.99965	0.99940	0.99964	0.99868

less than $0.0008\,k_B$ at the dense state. Discarding none, one quarter, or one half of production changes estimates by up to about $0.0045\,k_B$, comparable to some composition effects. The available data therefore support the large cutoff correction more securely than a precise ranking of closely spaced mixtures. Additional points below $\rho^* = 0.01$ and longer independent runs would test the remaining uncertainty.

D. Component Count, Mixing Entropy, and the Divalent-Ion Question

At fixed T, ρ, and equal particle masses, the difference in total entropy between two compositions is

$$\Delta s/k_B = \Delta[-\sum_i x_i \ln x_i] + \Delta s_{\rm ex}/k_B. \tag{33}$$

The ideal mixing contributions are 0.26856 (3C), 0.46390 (either 4C), and 0.65642 (5C). Thus 3C→5C increases ideal mixing entropy by $0.38787\,k_B$ per particle, while the reported MD excess entropy decreases by $0.00895\,k_B$. The resulting total entropy increase, approximately $0.37892\,k_B$, cannot be inserted into Eq. (1). A hypothetical split of the solvent into mechanically identical labels would leave forces, excess entropy, and diffusion unchanged while adding a formal mixing contribution if the labels are counted as distinguishable components.

Moreover, 3C→5C replaces solvent with monovalent ions: the total ion fraction rises from 0.06 to 0.16, and I from 0.12 to 0.17, while $x_{+2} = x_{-2} = 0.03$ remains fixed. This is a change in interactions and screening as well as component count. Similarly, the two 4C systems have identical mixing entropy; their differences arise from dispersion interactions. Under geometric mixing, changing ε_2 also changes solvent–ion cross interactions. Neither comparison isolates a purely entropic effect of component multiplicity.

E. Species-Resolved Partial Molar Entropies and Electrostriction

To establish how individual mixture components contribute to the macroscopic excess entropy, we evaluate the *partial molar excess entropy* from HNC theory. In solution thermodynamics, the partial molar excess entropy of component i at constant pressure is defined by the temperature derivative of its excess chemical potential:

$$\bar{S}^{\rm ex}_{i,P} \equiv \left(\frac{\partial S_{\rm ex}}{\partial N_i}\right)_{T,P,N_{j\neq i}} = -\left(\frac{\partial \mu^{\rm ex}_i}{\partial T}\right)_{P,\{x_k\}}. \tag{34}$$

At constant volume (or fixed overall density ρ), the corresponding canonical partial excess entropy is

$$s^{\rm ex}_{i,V} \equiv \left(\frac{\partial S_{\rm ex}}{\partial N_i}\right)_{T,V,N_{j\neq i}} = -\left(\frac{\partial \mu^{\rm ex}_i}{\partial T}\right)_{V,\{x_k\}}. \tag{35}$$

Using the thermal pressure coefficient $\gamma_V = (\partial P/\partial T)_V$ and the isothermal bulk modulus $(\partial P/\partial \rho)_T$, these quantities are related via the equation of state:

$$\bar{S}^{\rm ex}_{i,P} = s^{\rm ex}_{i,V} + \frac{\gamma_V}{(\partial P/\partial \rho)_T}\left(\frac{\partial \mu^{\rm ex}_i}{\partial \rho}\right)_T, \tag{36}$$

while the partial molar volume of species i is given by

$$\bar{V}_i = \left(\frac{\partial \mu_i}{\partial P}\right)_{T,\{x_k\}} = \frac{k_B T/\rho + (\partial \mu_i^{\mathrm{ex}}/\partial \rho)_T}{(\partial P/\partial \rho)_T}, \tag{37}$$

which should satisfy the volume additivity condition $\sum_{i=1}^{p} x_i \bar{V}_i = 1/\rho$.

Table III compiles the species-resolved thermodynamic properties evaluated at $\rho^* = 0.70$. Several key physical insights emerge:

1. **Solvent Dispersion Modulation**: In System 4C with enhanced dispersion ($\varepsilon_2 = 1.25$), Solvent 2 exhibits a strongly negative partial molar excess entropy ($\bar{S}_{i,P}^{\mathrm{ex}} = -1.580\, k_B$), reflecting rigid energetic caging. In contrast, reducing the dispersion depth to $\varepsilon_2 = 0.75$ triggers a pronounced entropic surge: $\bar{S}_{i,P}^{\mathrm{ex}}$ rises to $-1.115\, k_B$ ($\Delta \bar{S} = +0.465\, k_B$), accompanied by an expansion in partial molar volume ($\bar{V}_2 = 1.471\,\sigma^3$ vs $1.416\,\sigma^3$). The weakened dispersion dramatically expands the local configuration space available to the secondary solvent.

2. **Charge Hierarchy and Electrostriction**: Across all mixtures, ionic species exhibit systematically more negative partial molar entropies than neutral solvent molecules. In System 5C, a clear monotonic ordering follows the valence:

$$\begin{aligned} \bar{S}_{\mathrm{solv}}^{\mathrm{ex}}(-1.384\, k_B) &> \bar{S}_{|z|=1}^{\mathrm{ex}}(-1.445\, k_B) \\ &> \bar{S}_{|z|=2}^{\mathrm{ex}}(-1.636\, k_B). \end{aligned} \tag{38}$$

   This hierarchy directly reflects electrostriction: the intense Coulomb field of divalent counterions contracts the surrounding coordination shell, compressing the partial molar volume ($\bar{V}_{|z|=2} = 1.408\,\sigma^3 < \bar{V}_{|z|=1} = 1.427\,\sigma^3 < \bar{V}_{\mathrm{solv}} = 1.434\,\sigma^3$) and is consistent with changes in local translational confinement. These spherical particles have no rotational degrees of freedom.

3. **Concordance with Structural Pair Entropy**: As seen in Table III, the thermodynamic partial molar excess entropy $\bar{S}_{i,P}^{\mathrm{ex}}$ shows a similar qualitative species ordering to the Green-Wallace structural pair entropy $s_i^{(2)}$. These quantities have different definitions; their relationship and the bulk weighted-sum checks require further verification.

F. Two-Phase Thermodynamics (2PT) Analysis: Species-Resolved Decomposition across Multicomponent Mixtures

To provide an independent simulation-based decomposition of entropy and assess how the 2PT method performs across multicomponent electrolytes, we apply the 2PT framework [Eqs. (20)–(28)] to all four simulated fluids (Systems 3C, 4C with $\varepsilon_2 = 1.25$, 4C with $\varepsilon_2 = 0.75$, and 5C at $\rho^* = 0.70, T^* = 2.00$).

Figure 4 displays the comparative dynamic spectra across the four mixtures. In all components, the VACF [Fig. 4(a)–(b)] exhibits rapid initial decorrelation followed by a negative backscattering minimum around $t^* \approx 0.2$–$0.4\,\tau$, characteristic of cage rebound collisions in dense liquid states, before decaying to zero by $t^* \approx 2.5\,\tau$. Integrating the species VACFs yields Green-Kubo self-diffusion coefficients D_{GK}. As compiled in Table IV, D_{GK} matches the long-time Einstein self-diffusion coefficients D_{MSD} within 0.2%–5.1% for the majority solvent species and the net fluid, and within 0.1%–1.1% for most pooled ionic estimates, with an 11.2% discrepancy for the strongly attractive 4C divalent pool.

Figure 4(c)–(d) displays the corresponding vibrational densities of states $S_i(\nu)$. The spectra exhibit the characteristic dual-mode structure of dense liquids: a finite zero-frequency value $S(0)$ corresponding to diffusive escapes, and a broad solid-like vibrational band peaking around $\nu^* \approx 2.0$–$3.5\,\tau^{-1}$ representing interparticle rattling. Table IV compiles the complete 2PT decomposition parameters and species excess entropies alongside both Einstein (D_{MSD}) and Green-Kubo (D_{GK}) self-diffusion coefficients and the canonical partial excess entropies $s_{i,V}^{\mathrm{ex}}$ from HNC theory.

The spectral decompositions in Table IV show reduced secondary-solvent mobility for stronger attraction and broadly similar divalent-ion Green–Kubo diffusivities across the four systems. The numerical entropy columns are retained as outputs of the current implementation, but their thermodynamic interpretation remains provisional. A species contribution assigned by 2PT is not, by construction, the temperature derivative of a chemical potential. In particular, the 3C weighted 2PT value ($-5.376\, k_B$) differs substantially from corrected MD TI ($-1.9669\, k_B$). At the same equilibrium state entropy is a state function; an ensemble change alone does not explain this discrepancy. The ideal-mixture reference, spectral normalization, frequency convention, mode weights, and chemical-potential derivatives must be checked before interpreting the numerical proximity to the HNC derivative values as validation. With the VACF normalization in Eq. (20), the prefactor in Eq. (21) also needs checking: as written, its integrated spectral weight is one rather than three.

The reported salt-averaged Green–Kubo values are 0.2038, 0.2017, 0.2057, and 0.2037 for 3C, strongly attractive 4C, weakly attractive 4C, and 5C, respectively. Their full spread is only about 2%. By comparison, Table II gives ion MSD diffusivities from 0.1666 to 0.2289. The second MSD dataset in Table IV gives 0.2027, 0.1813, 0.2072, and 0.2036; the weakly attractive 4C ion slowdown in the first dataset is not reproduced. The 4C strongly attractive salt MSD and Green–Kubo values differ by about 11%. Furthermore, the listed pooled-ion MSD values need a common-frame additivity check against the equally weighted cation and anion values.

Charge-inversion symmetry of this model requires equal ensemble-averaged cation and anion self-diffusivities of

TABLE III. Species-resolved partial molar excess properties at $\rho^* = 0.70, T^* = 2.00$ ($\beta^* = 0.50, \Gamma = 1.0$) from HNC integral equation theory: excess chemical potential $\mu_i^{\rm ex}/(k_BT)$, partial molar volume $\bar{V}_i$, constant-volume partial excess entropy $s_{i,V}^{\rm ex}$, constant-pressure partial molar excess entropy $\bar{S}_{i,P}^{\rm ex}$, and structural Green-Wallace pair entropy $s_i^{(2)}$.

System	Component (i)	x_i	$\mu_i^{\rm ex}/(k_BT)$	$\bar{V}_i\,[\sigma^3]$	$s_{i,V}^{\rm ex}/k_B$	$\bar{S}_{i,P}^{\rm ex}/k_B$	$s_i^{(2)}/k_B$
3C ($I = 0.12$)	Solvent ($\varepsilon = 1.00$)	0.94	2.6031	1.445	−5.268	−1.363	−1.607
	Cation ($z = +2$)	0.03	2.0058	1.419	−5.441	−1.610	−1.729
	Anion ($z = -2$)	0.03	2.0058	1.419	−5.441	−1.610	−1.720
4C ($\varepsilon_2 = 1.25$)	Solvent 1 ($\varepsilon_1 = 1.00$)	0.89	2.5906	1.446	−5.281	−1.355	−1.610
	Solvent 2 ($\varepsilon_2 = 1.25$)	0.05	2.1808	1.416	−5.416	−1.580	−1.690
	Cation ($z = +2$)	0.03	1.9933	1.421	−5.453	−1.602	−1.740
	Anion ($z = -2$)	0.03	1.9933	1.421	−5.453	−1.602	−1.746
4C ($\varepsilon_2 = 0.75$)	Solvent 1 ($\varepsilon_1 = 1.00$)	0.89	2.6147	1.443	−5.250	−1.363	−1.605
	Solvent 2 ($\varepsilon_2 = 0.75$)	0.05	3.0487	1.471	−5.084	**−1.115**	**−1.558**
	Cation ($z = +2$)	0.03	2.0174	1.418	−5.422	−1.609	−1.744
	Anion ($z = -2$)	0.03	2.0174	1.418	−5.422	−1.609	−1.739
5C ($I = 0.17$)	Solvent	0.84	2.6021	1.434	−5.267	−1.384	−1.601
	Cation ($z = +1$)	0.05	2.4380	1.427	−5.308	−1.445	−1.645
	Anion ($z = -1$)	0.05	2.4380	1.427	−5.308	−1.445	−1.654
	Cation ($z = +2$)	0.03	1.9421	1.408	−5.442	−1.636	−1.753
	Anion ($z = -2$)	0.03	1.9421	1.408	−5.442	−1.636	−1.751

the same valence. Their finite-trajectory differences are a useful convergence diagnostic. Pooling improves sampling but neither guarantees cancellation of error nor measures the diffusion of an electrically neutral bound pair. Likewise, weak sensitivity of a computed 2PT entropy to a varied diffusion input does not demonstrate convergence of the diffusion coefficient or establish a physical failure of Rosenfeld scaling.

G. Transport Dynamics and Solvent-Mediated Diffusion

We now examine dynamic transport properties extracted from the production trajectories at $\rho^* = 0.70$. Figure 6 displays the center-of-mass corrected mean-squared displacements $\langle|\Delta\mathbf{r}(t)|^2\rangle$ for the net fluid and component species.

The mean MSDs are nearly linear over the chosen fitting windows ($R^2 \gtrsim 0.999$). This measures linearity of correlated MSD samples and is distinct from uncertainty in the long-time diffusion coefficient or goodness of an entropy-scaling fit. For the net fluid, the self-diffusion coefficient increases as solvent attraction weakens:

$$D_{\rm net}(\varepsilon_2 = 1.25) = 0.2149 \rightarrow D_{\rm net}(\varepsilon_2 = 0.75) = 0.2194. \tag{39}$$

Examining the component solvent species in the 4C mixtures reveals the microscopic mechanism governing this acceleration:

- In the more attractive fluid ($\varepsilon_2 = 1.25$), Solvent 2 is strongly trapped in local energy wells, diffusing slower than Solvent 1 ($D_{\rm solv2} = 0.2032$ vs $D_{\rm solv1} = 0.2167$; 0.2093 vs 0.2206 in Table IV).
- In the less attractive fluid ($\varepsilon_2 = 0.75$), Solvent 2 experiences reduced energetic resistance, escaping coordination shells and diffusing markedly faster than Solvent 1 ($D_{\rm solv2} = 0.2348$ vs $D_{\rm solv1} = 0.2216$; 0.2635 vs 0.2183 in Table IV).

H. Evidence and Limits of Rosenfeld Excess Entropy Scaling

For the three systems with $I = 0.12$, an unweighted fit of $\ln D_{\rm net}^*$ to corrected MD $s_{\rm ex}/k_B$ gives $B = 1.89$, $R^2 \simeq 0.69$, versus $B = 1.05$, $R^2 \simeq 0.39$ before correction. HNC entropies still give $B = 1.40$, $R^2 \simeq 0.78$. Including 5C in the corrected MD fit yields $B \simeq 0.54$, $R^2 \simeq 0.10$. The cutoff-only correction, retaining the original estimator, gives $B = 1.23$, $R^2 = 0.54$ for the solvent series; the further changes reflect the pressure estimator and screened integration. Thus the corrected solvent-series association improves, but a common four-mixture exponential is unsupported. Three closely spaced points and entropy uncertainties comparable to their spacing preclude a robust species-independent law. These descriptive fits do not incorporate uncertainty in diffusion or errors in the entropy coordinate.

The new sweep resolves the changing-LJ-Hamiltonian problem. The separate electrostatic real-space cutoffs only change the PPPM splitting at the specified accuracy. A screened dilute-limit treatment and block-based sampling estimates are now included; systematic uncertainty from the unsampled region below $\rho^* = 0.01$ remains. The large MSD-fit R^2 values in Table II continue to describe linearity rather than entropy scaling.

A mean-diffusion correlation need not imply a common species relation. The mean coefficient is dominated by solvent, which constitutes 84–94% of the particles; the

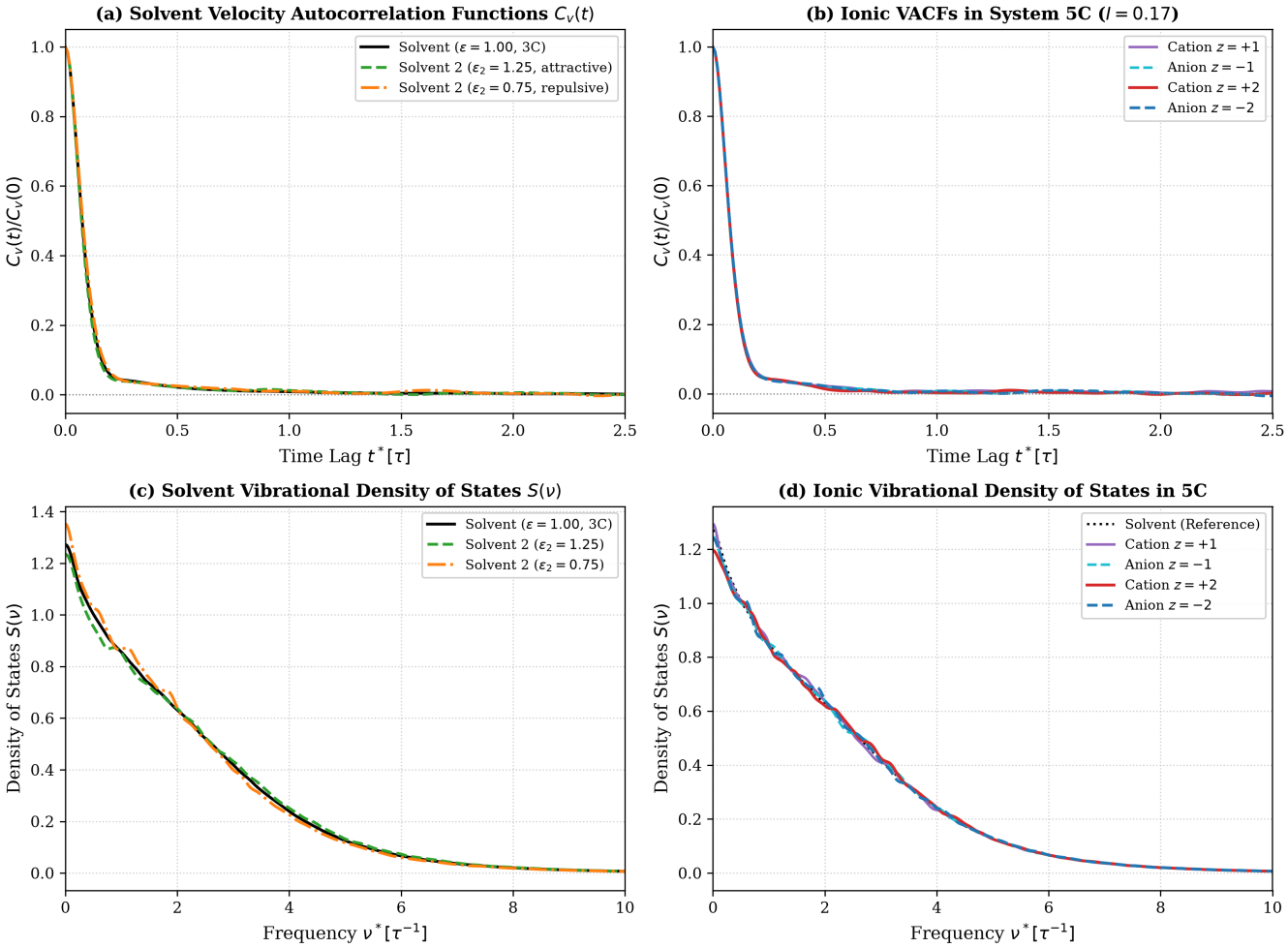


FIG. 4. Two-Phase Thermodynamics (2PT) dynamic spectra at $\rho^* = 0.70, T^* = 2.00$: (a) Normalized solvent velocity autocorrelation functions $C_v(t)/C_v(0)$, contrasting the baseline solvent ($\varepsilon = 1.00$, 3C), attractive secondary solvent ($\varepsilon_2 = 1.25$, 4C), and weakly attractive secondary solvent ($\varepsilon_2 = 0.75$, 4C); (b) Ionic VACFs in System 5C ($I = 0.17$), illustrating deeper backscattering wells for divalent counterions ($|z| = 2$) relative to monovalent ions ($|z| = 1$); (c) Vibrational densities of states $S(\nu)$ for the solvent species, showing the elevated zero-frequency diffusive peak $S(0)$ and softened vibrational band for $\varepsilon_2 = 0.75$; (d) Ionic vibrational densities of states in System 5C, highlighting the electrostrictive blue-shift of solid-like rattling bands for multivalent counterions.

divalent ions have only 6% weight. Even if the bulk relation held exactly, it would constrain a weighted sum rather than each D_i. For a *fixed species prefactor* and positive slope, Rosenfeld scaling would predict

$$\Delta \ln D_i^* = B_i \Delta s_{\mathrm{ex}}/k_B. \tag{40}$$

Thus the negative 3C→5C excess-entropy change would predict a slight decrease, not the approximately 6.8% ion increase in Table II. However, Table IV gives essentially no 3C→5C enhancement: approximately 0.4% by MSD and −0.05% by Green–Kubo. The magnitude and even ordering of ion trends are consequently unresolved.

A useful complementary structural descriptor is the species pair contribution

$$s_i^{(2)} = -2\pi\rho \sum_j x_j \int_0^\infty [g_{ij} \ln g_{ij} - (g_{ij} - 1)] r^2 \, dr, \tag{41}$$

which satisfies $s^{(2)} = \sum_i x_i s_i^{(2)}$. It is distinct from a thermodynamic partial molar entropy. Refitting the six solvent points embedded in the original figure generator yields $B \simeq 1.03$, $R^2 \simeq 0.87$, rather than the previously stated 1.05 and 0.91. These constants include a 3C mean diffusivity used as a solvent proxy and partial pair entropies whose extraction must be documented and checked against the weighted sum. They suggest a solvent association, not a universal species law. Generalized tracer entropy scaling introduces interaction- and composition-dependent kinetic information and can itself

TABLE IV. Two-Phase Thermodynamics (2PT) analysis across all four multicomponent mixtures at $\rho^* = 0.70, T^* = 2.00$ evaluated from extended $300\,\tau$ concurrent production trajectories (30 000 velocity frames): simulated Einstein self-diffusion from mean-squared displacements D_{MSD}, Green-Kubo self-diffusion D_{GK}, zero-frequency DoS $S(0)$, fluidicity f, diffusive excess entropy $s_{\mathrm{ex}}^{\mathrm{diff}}$, solid vibrational excess entropy $s_{\mathrm{ex}}^{\mathrm{solid}}$, and reported 2PT species excess-entropy contribution $s_{i,V}^{\mathrm{2PT}}$ compared against HNC constant-volume partial excess entropy $s_{i,V}^{\mathrm{ex}}$. For ionic species, pooled self-diffusion estimates ($|z| = 1$ and $|z| = 2$) are included. Pooling does not measure neutral-pair diffusion. Entropy definitions and normalization remain provisional, as discussed in the text.

System	Component (i)	x_i	D_{MSD}	D_{GK}	$S(0)$	f	$s_{\mathrm{ex}}^{\mathrm{diff}}/k_B$	$s_{\mathrm{ex}}^{\mathrm{solid}}/k_B$	$s_{i,V}^{\mathrm{2PT}}/k_B$	HNC $s_{i,V}^{\mathrm{ex}}/k_B$
3C ($I = 0.12$)	Solvent ($\varepsilon = 1.00$)	0.94	0.2307	0.2189	1.2684	0.3593	−0.684	−4.687	**−5.371**	−5.268
	Cation ($z = +2$)	0.03	0.2231	0.2052	1.2014	0.3505	−0.719	−4.734	**−5.453**	−5.441
	Anion ($z = -2$)	0.03	0.1793	0.2023	1.1991	0.3500	−0.721	−4.744	**−5.465**	−5.441
	Divalent Salt ($\|z\| = 2$)	0.06	0.2027	0.2038	1.2003	0.3503	−0.720	−4.739	*-5.459*	−5.441
	Net Fluid	1.00	0.2292	0.2180	1.2644	0.3588	−0.686	−4.690	*-5.376*	—
	Mixture ($\sum x_i s_i$)	1.00	—	—	—	—	—	—	**−5.376**	**−5.278**
4C ($\varepsilon_2 = 1.25$)	Solvent 1 ($\varepsilon_1 = 1.00$)	0.89	0.2206	0.2193	1.2695	0.3595	−0.684	−4.689	**−5.372**	−5.281
	Solvent 2 ($\varepsilon_2 = 1.25$)	0.05	0.2093	0.2076	1.2183	0.3526	−0.710	−4.785	**−5.495**	−5.416
	Cation ($z = +2$)	0.03	0.2038	0.1952	1.1897	0.3486	−0.727	−4.753	**−5.479**	−5.453
	Anion ($z = -2$)	0.03	0.1532	0.2082	1.2324	0.3542	−0.704	−4.735	**−5.439**	−5.453
	Divalent Salt ($\|z\| = 2$)	0.06	0.1813	0.2017	1.2110	0.3514	−0.715	−4.744	*-5.459*	−5.453
	Net Fluid	1.00	0.2182	0.2177	1.2635	0.3587	−0.687	−4.697	*-5.383*	—
	Mixture ($\sum x_i s_i$)	1.00	—	—	—	—	—	—	**−5.384**	**−5.298**
4C ($\varepsilon_2 = 0.75$)	Solvent 1 ($\varepsilon_1 = 1.00$)	0.89	0.2183	0.2196	1.2751	0.3602	−0.681	−4.681	**−5.362**	−5.250
	Solvent 2 ($\varepsilon_2 = 0.75$)	0.05	0.2635	0.2288	1.3350	0.3682	−0.651	−4.563	**−5.214**	**−5.084**
	Cation ($z = +2$)	0.03	0.2300	0.2054	1.2370	0.3549	−0.701	−4.711	**−5.412**	−5.422
	Anion ($z = -2$)	0.03	0.1797	0.2060	1.2191	0.3533	−0.708	−4.715	**−5.423**	−5.422
	Divalent Salt ($\|z\| = 2$)	0.06	0.2072	0.2057	1.2280	0.3541	−0.705	−4.713	*-5.418*	−5.422
	Net Fluid	1.00	0.2199	0.2193	1.2753	0.3602	−0.681	−4.677	*-5.358*	—
	Mixture ($\sum x_i s_i$)	1.00	—	—	—	—	—	—	**−5.358**	**−5.252**
5C ($I = 0.17$)	Solvent	0.84	0.2249	0.2188	1.2671	0.3592	−0.685	−4.688	**−5.373**	−5.267
	Cation ($z = +1$)	0.05	0.2041	0.2213	1.2429	0.3562	−0.696	−4.704	**−5.400**	−5.308
	Cation ($z = +2$)	0.03	0.1790	0.2054	1.2048	0.3510	−0.717	−4.730	**−5.447**	−5.442
	Anion ($z = -1$)	0.05	0.2326	0.2229	1.2669	0.3591	−0.685	−4.696	**−5.381**	−5.308
	Anion ($z = -2$)	0.03	0.2201	0.2019	1.1928	0.3492	−0.724	−4.738	**−5.462**	−5.442
	Monovalent Salt ($\|z\| = 1$)	0.10	0.2196	0.2221	1.2549	0.3577	−0.691	−4.700	*-5.391*	−5.308
	Divalent Salt ($\|z\| = 2$)	0.06	0.2036	0.2037	1.1988	0.3501	−0.720	−4.734	*-5.454*	−5.442
	Net Fluid	1.00	0.2233	0.2182	1.2617	0.3585	−0.688	−4.692	*-5.379*	—
	Mixture ($\sum x_i s_i$)	1.00	—	—	—	—	—	—	**−5.379**	**−5.282**

fail when component mobilities decouple [9]; applicability to screened Coulomb mixtures requires a separate test.

Added monovalent salt could alter divalent-ion association, exchange kinetics, and the ionic atmosphere, potentially increasing mobility even if bulk excess entropy decreases. Conversely, stronger clustering or friction could reduce it. These are competing mechanisms, not consequences of ideal mixing entropy. The model's neutral solvent has no explicit dielectric response, and the Coulomb prefactor is fixed; changing LJ attraction does not directly change dielectric shielding. Static contact peaks alone cannot establish pair lifetimes, populations, or their contribution to diffusion.

I. Further Checks

The first priority is a common-frame transport analysis of independent trajectories, with multiple time origins, block or replica confidence intervals, fitting-window stability, and matched MSD/VACF estimators. Both signs of each valence should agree within uncertainty. Verify $D_{\mathrm{net}} = \sum_i x_i D_i$ and the pooled divalent average before attributing differences to ion association. The fixed-LJ integration is now complete; extend the dilute grid below $\rho^* = 0.01$ and obtain independent trajectories to test the remaining integration and sampling uncertainties.

Next, introduce an otherwise identical 4C solvent with $\varepsilon_2 = 1$ as a label-splitting control. Scan added 1:1 salt at fixed divalent fraction, total density, and temperature, and compare with a separate constant-I replacement series. Constant I alone does not fix ion fraction or local chemistry; report both. Measure ion coordination numbers, associated fractions, cluster sizes, continuous and intermittent pair lifetimes, and solvent exchange, using consistent shell definitions and cutoff-sensitivity checks. This would test whether an ion speedup accompanies shorter-lived association.

Increase the divalent fraction symmetrically, for example from $x_{+2} = x_{-2} = 0.03$ to 0.06 and 0.10, replacing solvent and checking phase stability. This increases ion statistics and the ionic contribution to bulk entropy, while

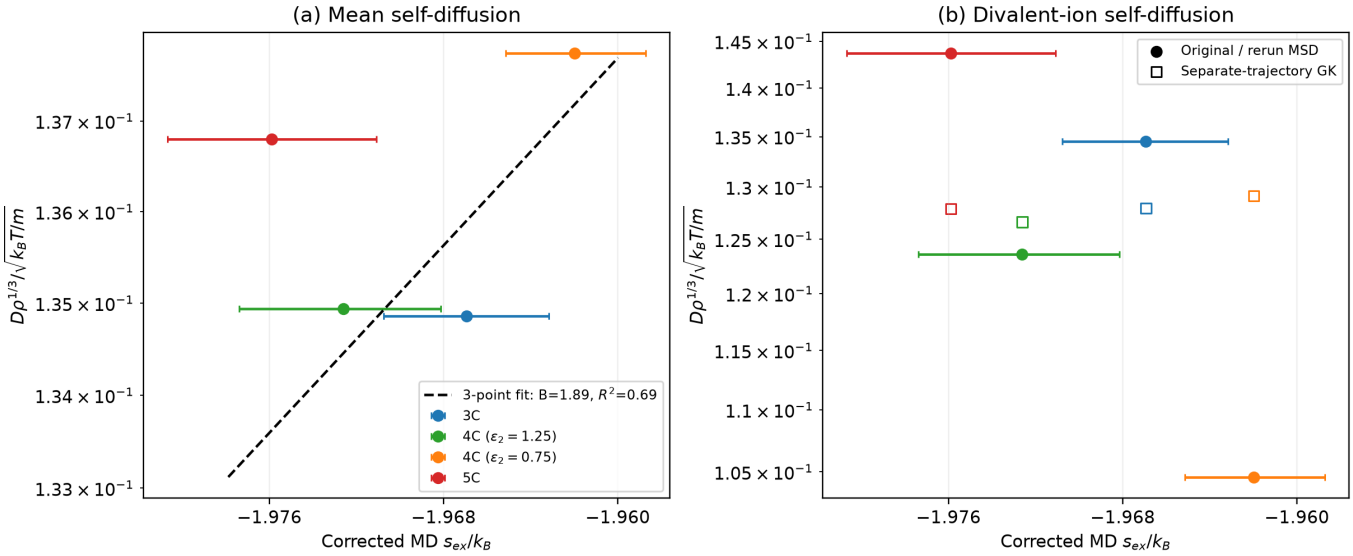


FIG. 5. Corrected entropy–diffusion comparison at $\rho^* = 0.70$, $T^* = 2.00$. (a) Mean reduced self-diffusion versus corrected MD excess entropy: the solvent-series fit has $B = 1.89$, $R^2 = 0.69$. Horizontal bars show conditional 95% block-bootstrap sampling intervals for entropy. (b) Divalent-ion MSD estimates (filled circles, unchanged by the reruns) and separate-trajectory Green–Kubo values (open squares), against the same corrected entropies. Diffusion uncertainty is not represented by these horizontal bars.

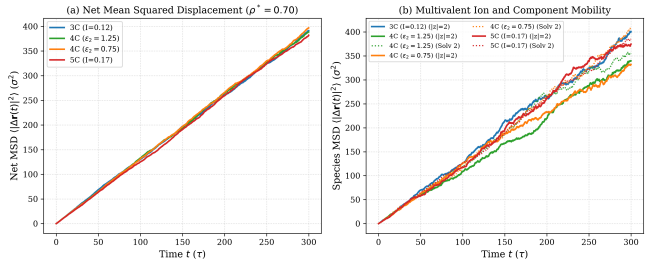


FIG. 6. Center-of-mass corrected mean-squared displacement (MSD) as a function of time t^* at $\rho^* = 0.70, T^* = 2.00$: (a) Net fluid MSD; (b) Multivalent ion ($|z| = 2$) and secondary solvent species MSD across all systems.

also changing screening and association. Independently vary ion-to-solvent diameter ratios, for example 0.8, 1.0, and 1.2 at equal cation/anion diameters and fixed masses, with packing fraction controlled or reported. Particle-size changes affect contact electrostatic coupling, so include controls matching $|z_i z_j|\lambda_B/\sigma_{ij}$ when separating packing and association effects.

Simulation size should be tested separately from particle size. Compare, for example, $N = 2000$, 4000, and 8000 at identical composition and density, extrapolating D_i versus $1/L$ and checking the cubic-box hydrodynamic correction $D_i(\infty) \simeq D_i(L) + 2.837297 k_B T/(6\pi\eta L)$ with independently measured mixture viscosity [10]. Extend sampling beyond measured association lifetimes and test thermostat sensitivity. Finally, obtain diffusion over several densities and temperatures, beyond the single dense state used here. Fit and validate mean and species scaling separately, using held-out states and uncertainty in entropy as well as diffusion. A stringent comparison is between states with matched excess entropy but different compositions.

V. CONCLUSIONS

The principal question is whether increasing mixture complexity raises entropy in a way that accelerates the main $+2/-2$ species. The present data support the following conclusions, in order of relevance to that question.

1. **Component count does not supply a Rosenfeld driving variable.** The larger total entropy of 5C is primarily ideal mixing entropy, which is subtracted from s_{ex}. Its reported excess entropy is slightly more negative than that of 3C. The current compositions therefore do not demonstrate a component-count-induced increase in excess entropy or ion diffusion.

2. **Solvent dispersion gives the clearest observed trend.** Weakening the secondary solvent attraction raises the reported excess entropy and increases mean self-diffusion by approximately 2.1% between the two 4C mixtures. The mean coefficient is solvent dominated. Its positive Rosenfeld slope is suggestive, but the three-point MD entropy fit ($R^2 \simeq 0.69$) and weak four-mixture fit ($R^2 \simeq 0.10$) preclude a universal relation.

3. **Divalent-ion scaling remains unresolved.** The original MSD dataset does not track bulk excess

entropy, but its large ion changes are not reproduced by the second dataset's nearly constant Green–Kubo values. Neither an ion speedup due to additional components nor a definitive physical breakdown of species entropy scaling is established. Association and screening are plausible mechanisms requiring independent structural and kinetic evidence.

4. **The thermodynamic calculations provide supporting structure, with limits.** HNC free-energy differentiation agrees with its virial route, and pair entropy accounts for approximately 76.1% of the HNC excess entropy at the dense state. The fixed-LJ reruns remove about half of the previous absolute entropy discrepancy, but internal consistency does not remove the remaining HNC–MD difference or the need to audit the reported partial-entropy and 2PT references and normalization.

The decisive next steps are reproducible common-frame ion diffusivities and verification of the unsampled dilute limit, followed by controlled salt addition, association-lifetime measurements, and independent particle-size, concentration, and box-size tests. These comparisons can determine whether mobility enhancement accompanies a genuine increase in excess entropy or instead reflects composition-specific changes in local friction.

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